

Failure Propagation in Multi-Agent LLM Systems: Characterization and Containment with Circuit Breakers

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Abstract

Large language models (LLMs) are increasingly deployed as compositional systems in which multiple agents interact to solve complex tasks. While such architectures enable modularity and improved capability, they also introduce a new class of failure modes: errors originating in one component can propagate through the system, leading to cascade failures.

We study failure propagation in multi-agent LLM pipelines and evaluate circuit breakers as a mechanism for containment. We introduce Cascade Failure Rate (CFR) and provide both simulation and real-world validation.

Our results show that circuit breakers substantially reduce cascade failures, but introduce a safety-availability tradeoff. Crucially, we show that this tradeoff is not inherent to circuit breaking itself, but emerges when fallback performance degrades under load. We further demonstrate that adaptive circuit breakers achieve strong reductions in catastrophic failures and yield increasing utility gains in high-risk regimes.

These findings suggest that reliability in compositional LLM systems is governed not only by component-level accuracy, but by the dynamics of failure propagation and the quality of fallback mechanisms.

1 Introduction

Multi-agent LLM systems are increasingly used for planning, tool use, and reasoning. These systems improve capability, but they also introduce new failure modes.

A key challenge is failure propagation: errors introduced in early stages can cascade through downstream components. Unlike isolated errors, these failures compound across stages.

Failure propagation scales with pipeline depth and dependency structure, making system reliability fundamentally different from component-level reliability. We therefore frame compositional LLM systems as distributed systems in which reliability depends on controlling error propagation, not only improving the behavior of individual components. Inspired by circuit breakers in distributed systems [2, 3], we study mechanisms that contain failures before they cascade.

Contributions. This paper makes five contributions:

- We formalize failure propagation in multi-agent LLM systems using Cascade Failure Rate (CFR).
- We introduce a simulation framework parameterized by intrinsic failure probability and propagation probability.
- We evaluate circuit breaker mechanisms for containment, including AI-guided and adaptive policies.

- We show that the safety–availability tradeoff is induced by degraded fallback performance rather than being inherent to circuit breaking.
- We demonstrate that adaptive policies provide increasing utility gains as the cost of catastrophic failure increases.

2 Theoretical Framework

Let p_u denote upstream failure probability and p_p denote propagation probability.

$$P(\text{final failure}) = p_u \cdot p_p \quad (1)$$

We define Cascade Failure Rate (CFR) as:

$$\text{CFR} = P(\text{final failure} \mid \text{upstream failure}) \quad (2)$$

For a pipeline of length L , a simple independent-propagation model yields:

$$\text{CFR}(L) = 1 - (1 - p_p)^L \quad (3)$$

which is monotonically increasing in L .

Implication. Even moderate propagation probabilities lead to high end-to-end failure rates in deeper pipelines. Circuit breakers reduce effective propagation by interrupting execution before failures cascade.

Utility. To study the tradeoff between containment and task completion, we also use a simple utility objective:

$$U = P(\text{success}) - c \cdot P(\text{catastrophic failure}) \quad (4)$$

where c is the cost assigned to catastrophic failure. This captures settings in which catastrophic outcomes are substantially more expensive than safe failure or intervention.

3 Method

We evaluate three configurations:

- **No Circuit Breaker**
- **AI-guided Circuit Breaker**
- **Adaptive Circuit Breaker**

Pipeline configuration. For real-world validation, we use a three-stage planner–executor–verifier pipeline. Each stage produces structured outputs that are consumed by downstream stages.

Simulation states. In simulation, each execution can be in one of three states:

- **Healthy:** normal execution
- **Recoverable:** degraded but potentially successful without intervention
- **Risky:** likely to produce catastrophic downstream failure if execution continues

Failure injection. Failures are injected according to upstream failure probability p_u , while propagation is governed by p_p .

Fallback behavior. When a breaker intervenes, execution switches to a fallback path. We model fallback success probability as degrading under load. Specifically, degradation strength is parameterized by α , so that more aggressive intervention becomes more expensive when the fallback path is itself stressed.

Failure definition. An upstream failure is defined as an intermediate output that is invalid, malformed, incomplete, or semantically inconsistent.

Circuit breaker policy. The AI-guided circuit breaker evaluates intermediate outputs using a secondary model-based risk signal and triggers when the risk score exceeds a fixed threshold.

The adaptive circuit breaker dynamically adjusts its intervention threshold based on estimated system risk. Intuitively, it becomes more conservative in high-propagation regimes, intervening earlier to contain risky states while still tolerating some recoverable errors.

Metrics. We report:

- Cascade Failure Rate (CFR)
- Success Rate
- Safe Failure Rate
- Breaker Activation Rate
- Utility

4 Experiments

4.1 Simulation

We simulate pipelines under varying values of p_u and p_p . The main sweeps vary:

- propagation probability p_p
- fallback degradation strength α
- catastrophe cost c
- fallback degradation model (linear subtractive, linear multiplicative, exponential)

Each configuration is evaluated using Monte Carlo simulation over 10^4 runs.

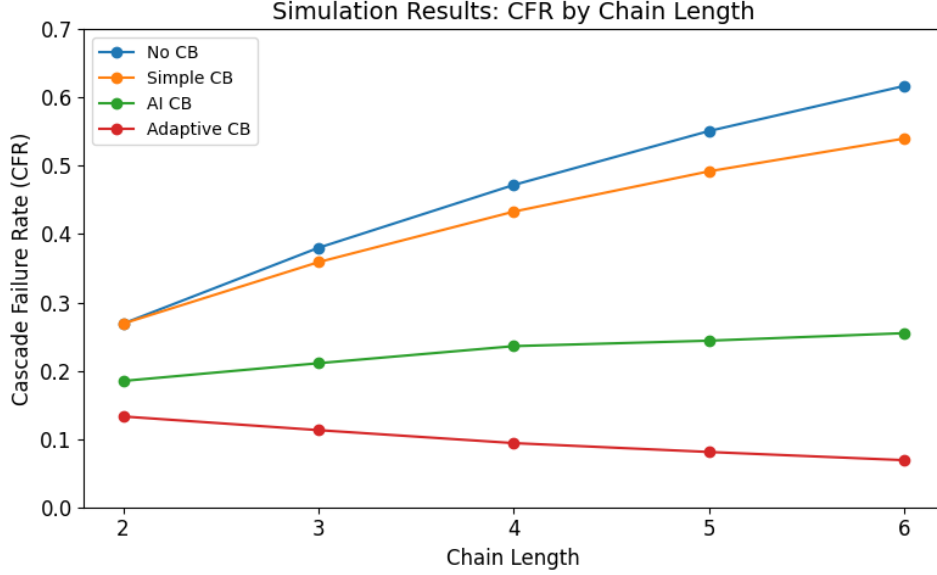


Figure 1: Cascade Failure Rate (CFR) increases with pipeline depth. Circuit breakers reduce effective propagation and flatten this growth.

4.2 Real-world API setup

We evaluate a planner-executor-verifier pipeline using a GPT-4-class LLM API. Each experiment consists of 50 independent runs on a synthetic task set requiring structured reasoning and multi-step execution.

Tasks involve generating intermediate structured outputs that are consumed by downstream stages. Failures are injected probabilistically at intermediate stages, and propagation is observed through to final task outcomes.

An upstream failure is defined as an intermediate output that is invalid, incomplete, or low-confidence. A cascade occurs when such a failure leads to final task failure.

5 Results

5.1 Simulation depth validates the propagation model

Figure 1 shows that CFR increases with pipeline depth, consistent with the theoretical model. This validates the core claim that failure propagation grows as systems become deeper and more compositional.

Circuit breakers reduce effective propagation, flattening the growth of CFR and lowering the system-level probability that an upstream failure reaches the final output.

5.2 Real-world validation confirms cascade containment

Table 1 and Figure 2 summarize the real-world API results.

Without circuit breakers, 62% of upstream failures propagate to final failure, demonstrating meaningful cascade effects in a practical multi-agent pipeline. Both AI-guided and adaptive breakers eliminate cascade failures entirely in this setting. However, the AI-guided breaker achieves slightly

Method	Cascade Rate	Success
No CB	0.62	0.38
AI CB	0.00	0.44
Adaptive CB	0.00	0.40

Table 1: Real-world API results averaged over 50 runs.

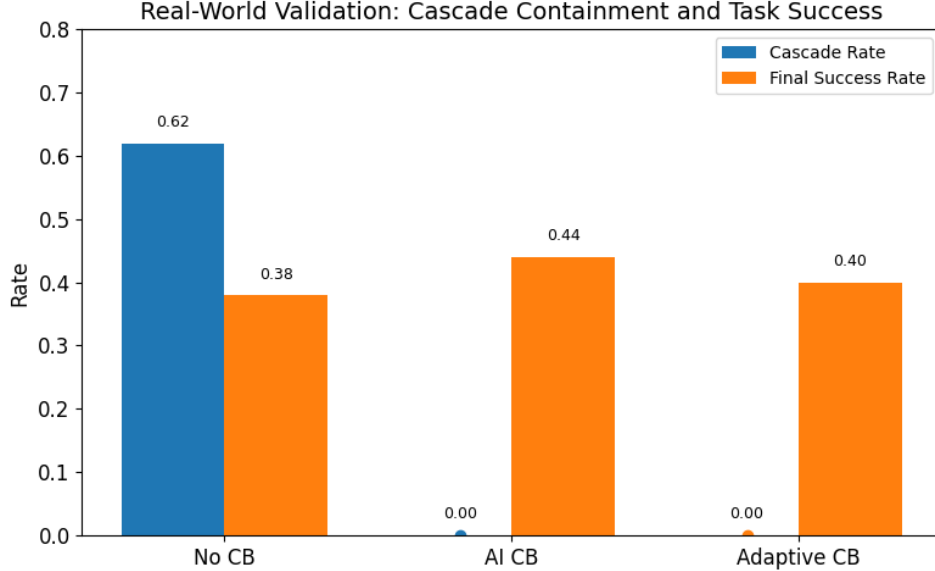


Figure 2: Real-world validation of circuit breaker policies in a planner-executor-verifier pipeline.

higher final success, indicating that selective intervention can outperform uniformly aggressive suppression.

5.3 Fallback degradation induces a safety-availability tradeoff

We next examine how fallback degradation changes the relative behavior of adaptive and AI-guided circuit breakers. Figure 3 plots the difference in success rate and catastrophic failure rate (CFR) between adaptive and AI-guided policies across increasing propagation probability p_p , for varying degradation strength α .

Across all settings, adaptive breakers consistently reduce catastrophic failures ($\Delta\text{CFR} < 0$), indicating improved safety. However, the impact on success rate is more nuanced. For weak degradation ($\alpha = 0.3$), adaptive breakers achieve both higher or comparable success and lower CFR across most regimes. As degradation strengthens ($\alpha = 0.5, 0.7$), a region emerges where adaptive breakers reduce CFR at the cost of lower success.

We refer to this regime as the *safety-availability tradeoff region*, where $\Delta\text{CFR} < 0$ and $\Delta\text{success} < 0$. Notably, this tradeoff is absent or minimal under weak degradation but expands as α increases. This demonstrates that the tradeoff is not inherent to circuit breaking itself, but rather induced by fallback quality.

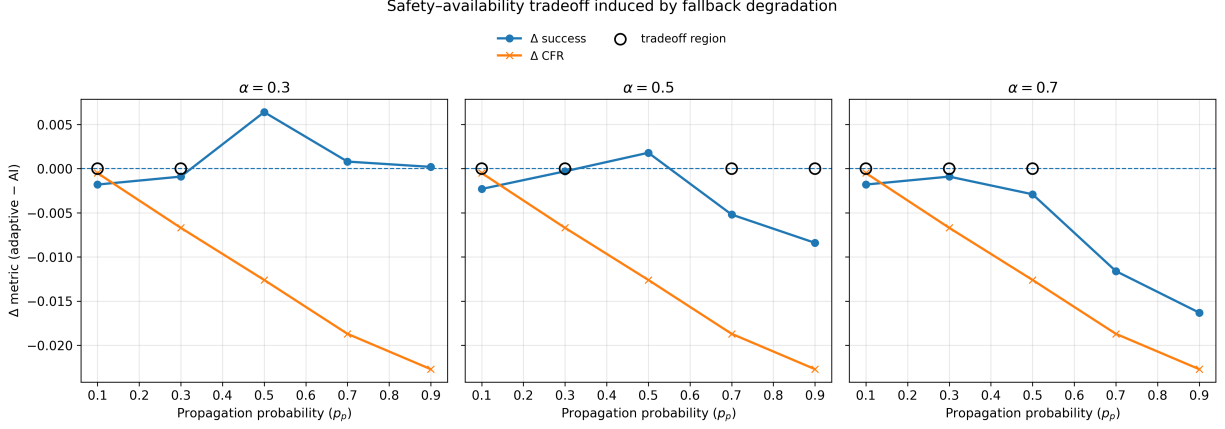


Figure 3: Safety-availability tradeoff induced by fallback degradation. Difference in success rate (Δ success) and catastrophic failure rate (Δ CFR) between adaptive and AI-guided breakers across propagation probability p_p , for varying degradation strength α . Adaptive breakers consistently reduce catastrophic failures. Under stronger degradation, a tradeoff region emerges where adaptive breakers reduce CFR at the cost of lower success.

5.4 Higher catastrophe cost favors adaptive breakers

Figure 4 evaluates utility under varying catastrophe costs.

Across all propagation probabilities, adaptive breakers achieve higher utility than AI-guided breakers, and this advantage increases monotonically with catastrophe cost. While differences in success rate remain modest, the consistent reduction in catastrophic failures yields substantial gains in expected utility when catastrophic outcomes are heavily penalized.

This result highlights an important decision-theoretic distinction: even when two policies have similar success rates, the one that better suppresses catastrophic failures becomes increasingly preferable in high-risk settings.

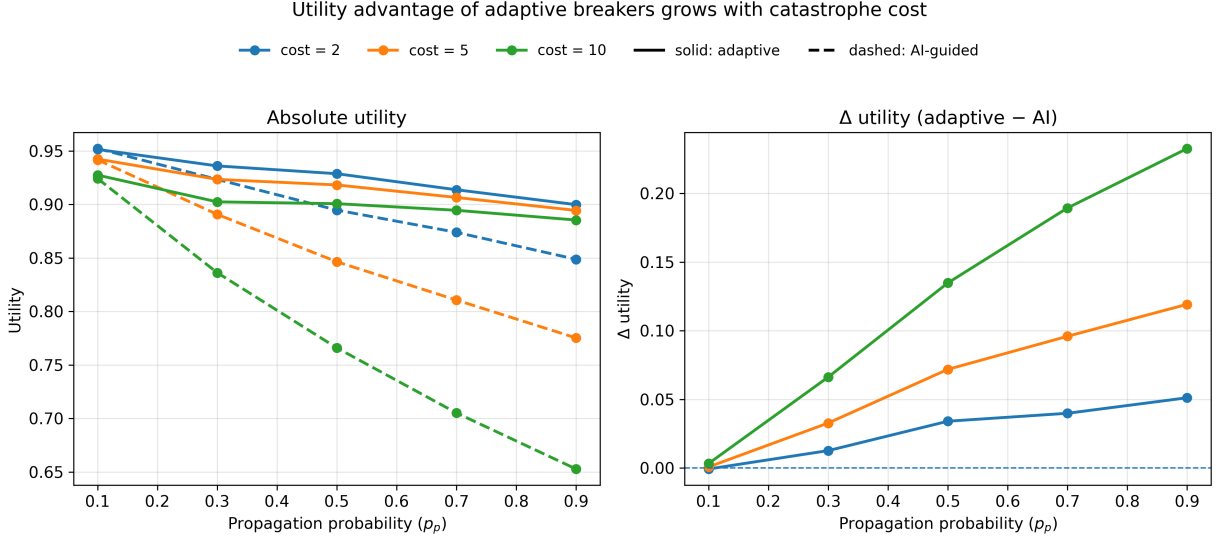


Figure 4: Utility advantage of adaptive breakers grows with catastrophe cost. Left: absolute utility of adaptive (solid) and AI-guided (dashed) breakers. Right: utility difference (Δ utility = adaptive – AI-guided). Adaptive breakers consistently outperform AI-guided breakers, with gains increasing monotonically with catastrophe cost and propagation probability.

5.5 Results are robust across degradation models

To ensure that our findings are not artifacts of a particular degradation formulation, we evaluate multiple fallback degradation models, including linear subtractive, linear multiplicative, and exponential decay. Figure 5 reports the corresponding differences in success and CFR.

Across all models, the qualitative behavior remains consistent: adaptive breakers reduce catastrophic failures across all regimes, while success-rate differences vary depending on degradation severity. The emergence of the safety–availability tradeoff under stronger degradation is observed across all models.

These results demonstrate that our conclusions are robust to the choice of degradation function, and that the observed tradeoffs arise from degradation itself rather than a particular modeling assumption.

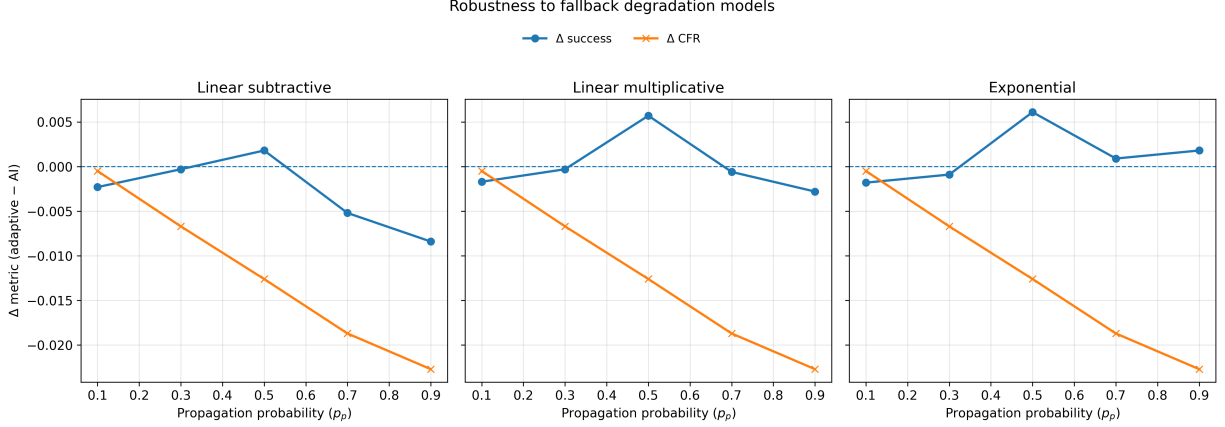


Figure 5: Robustness to fallback degradation models. Differences in success rate and catastrophic failure rate between adaptive and AI-guided breakers under multiple degradation models. Across all models, adaptive breakers consistently reduce catastrophic failures, while success varies with degradation strength.

6 Discussion

Failure propagation is a system-level phenomenon. Reliability depends on controlling propagation dynamics rather than only improving individual components.

Our results suggest that compositional LLM systems should be designed with explicit failure containment mechanisms. This shifts the focus from improving individual model outputs to managing system-level error dynamics.

Importantly, containment alone is not sufficient. The effectiveness of circuit breaking depends critically on fallback quality. When fallback performance degrades under load, a safety-availability tradeoff emerges: aggressive containment reduces catastrophic failures but may also suppress successful outcomes.

This suggests that optimal reliability mechanisms must jointly optimize containment and task success. Rather than minimizing failures alone, systems should account for the downstream impact of interventions on overall utility.

7 Related Work

Multi-agent systems such as AutoGPT [6] and BabyAGI [1] demonstrate compositional LLM architectures. Toolformer [4] and ReAct [8] explore tool use and reasoning.

Error propagation in machine learning pipelines has been studied in prior work [5], which highlights how small upstream issues can amplify downstream.

Recent work on LLM reliability has focused on reasoning robustness, self-consistency, and evaluation stability [7]. Our work complements these directions by focusing on system-level failure dynamics rather than only model-level output quality.

Distributed systems literature [2, 3] motivates circuit breaker design. Our work extends this framing to LLM systems, where failures are probabilistic and semantic rather than binary.

8 Limitations

We evaluate relatively simple pipelines and controlled failure injections. Real-world systems may exhibit richer dependency structures, feedback loops, and more heterogeneous fallback behavior than those captured in our simulation.

Our real-world validation is intentionally small-scale and should be interpreted as a proof of concept rather than a comprehensive benchmark.

9 Conclusion

We study failure propagation in multi-agent LLM systems and show that circuit breakers provide an effective mechanism for cascade containment. Beyond showing that containment matters, we identify a more nuanced result: the safety–availability tradeoff is induced by degraded fallback performance, rather than being inherent to circuit breaking itself.

Adaptive policies reduce catastrophic failures and become increasingly valuable as the cost of catastrophic outcomes rises. Together, these findings suggest that reliability in compositional LLM systems should be treated as a control problem over failure propagation and fallback quality.

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